MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP BCA (HONOURS)

FIFTH SEMESTER EXAMINATION (2024 ADMISSION ONWARDS)

MG5DSEBCA300: INTRODUCTION TO DATA SCIENCE

MODEL QUESTION PAPER

Duration: 2hrs Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Answer all questions.

Each question carries 2 marks.

1. What is Data Science? Mention any one need for Data Science. **[K] [CO1]**

2. List the steps involved in the Data Science Life Cycle? **[K] [CO1]**

3. Define data. Mention any two types of data. **[K] [CO2]**

4. What is Exploratory Data Analysis? State its importance. **[U] [CO3]**

5. What is meant by Data Science Modelling? **[K] [CO4]**

(5 × 2 = 10 Marks)

PART B

Answer any five questions.

Each question carries 6 marks.

6. Explain the evolution of Data Science and describe the different roles in Data Science. **[U] [CO1]**

7. Explain the applications of Data Science in various fields. Discuss the prerequisites required for learning Data Science. **[U] [CO1]**

8. Discuss any three technical skills required for working with data science. **[U] [CO1]**

9. Explain the basic features of R and describe R data types and operators with suitable examples. **[U] [CO2]**

10. Explain the different types of Exploratory Data Analysis with suitable examples. **[U] [CO3]**

11. Explain Univariate, Bivariate and Multivariate Analysis with suitable examples**. [U] [CO3]**

12. Explain K-fold Cross-Validation and Stratified Cross-Validation**. [U] [CO4]**

(5 × 6 = 30 Marks)

PART C

Answer any two questions.

Each question carries 15 marks.

13. Duscuss the major Data Pre-processing techniques used to prepare raw data for analysis. **[An] [CO2]**

14. Apply the steps of EDA to a given dataset. Illustrate how appropriate EDA techniques and tools can be used to understand the characteristics and patterns of the data. **[A] [CO3]**

15. Analyse the process of Linear Model Selection and Diagnostics. Explain the importance of Model Evaluation and Metrics in Data Science. **[An] [CO4]**

(2 × 15 = 30 Marks)